WORKED SOLUTION

STEP I 1988, Question 5

Pure • Series and Generating Functions

STEP I 1988, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q5

The key idea

Expand each factor as a geometric series (the binomial theorem with index -1) and multiply: c_m is then a finite geometric sum, which has a neat closed form. Once c_m^2 is written as a combination of three pure powers, $(a^2)^m$, $(ab)^m$ and $(b^2)^m$, the final series splits into three geometric series. The rest is careful algebra with fractions, and keeping track of where each series converges.

Solution

Finding c_m

By the binomial theorem with index -1,

$$(1-ax)^{-1} = \sum_{k=0}^{\infty} a^k x^k \quad (|ax|<1), \qquad (1-bx)^{-1} = \sum_{j=0}^{\infty} b^j x^j \quad (|bx|<1).$$

Since $0 < a < b$, the condition $b|x| < 1$ gives $a|x| < b|x| < 1$, so both expansions are valid. Both series converge absolutely, so we may multiply them and collect powers of x. The coefficient of x^m comes from the pairs $k + j = m$:

$$c_m = \sum_{k=0}^{m} a^k b^{m-k} = b^m + ab^{m-1} + \cdots + a^{m-1}b + a^m.$$

This is a geometric series with first term b^m, common ratio a/b and $m+1$ terms. Since $a \neq b$, its sum is

$$c_m = \frac{b^{m+1} - a^{m+1}}{b-a}.$$

(Equivalently, $(b-a)c_m = b^{m+1} - a^{m+1}$, because the sum telescopes when multiplied out.)

The formula for c_m^2

Squaring, and using $(b-a)^2 = (a-b)^2$,

$$c_m^2 = \frac{(b^{m+1} - a^{m+1})^2}{(b-a)^2} = \frac{a^{2m+2} - 2(ab)^{m+1} + b^{2m+2}}{(a-b)^2},$$

as required.

The series $\sum c_m^2 x^m$

The point of the previous formula is that each term is a constant times an mth power:

$$c_m^2 x^m = \frac{1}{(a-b)^2}\left(a^2\,(a^2x)^m - 2ab\,(abx)^m + b^2\,(b^2x)^m\right).$$

So $\sum c_m^2 x^m$ is a combination of three geometric series, with ratios a^2x, abx and b^2x.

? **Ask yourself** For which x do all three geometric series converge?

We need $|a^2x| < 1$, $|abx| < 1$ and $|b^2x| < 1$. Since $0 < a^2 < ab < b^2$, the last condition is the strongest and implies the other two. So all three converge exactly when $|x| < 1/b^2$. Then, summing each geometric series,

$$\sum_{m=0}^{\infty} c_m^2 x^m = \frac{1}{(a-b)^2}\left(\frac{a^2}{1-a^2x} - \frac{2ab}{1-abx} + \frac{b^2}{1-b^2x}\right).$$

Put everything over the common denominator $(1-abx)(1-a^2x)(1-b^2x)$. The numerator is

$$N = a^2(1-abx)(1-b^2x) - 2ab(1-a^2x)(1-b^2x) + b^2(1-a^2x)(1-abx).$$

We collect it by powers of x:

$$\begin{aligned}
\text{constant term:} \quad & a^2 - 2ab + b^2 = (a-b)^2, \\
\text{coefficient of } x: \quad & -a^2(ab+b^2) + 2ab(a^2+b^2) - b^2(a^2+ab) \\
& = a^3b - 2a^2b^2 + ab^3 = ab(a-b)^2, \\
\text{coefficient of } x^2: \quad & a^2 \cdot ab^3 - 2ab \cdot a^2b^2 + b^2 \cdot a^3b = 0.
\end{aligned}$$

So $N = (a-b)^2(1+abx)$. The factor $(a-b)^2$ cancels, which leaves

$$c_0^2 + c_1^2 x + c_2^2 x^2 + \cdots = \frac{1+abx}{(1-abx)(1-a^2x)(1-b^2x)} \qquad \text{for } -\frac{1}{b^2} < x < \frac{1}{b^2}.$$

The interval can't be any larger. At $x = \pm 1/b^2$ the terms satisfy

$$|c_m^2 x^m| = \frac{(b^{m+1} - a^{m+1})^2}{(b-a)^2\, b^{2m}} = \frac{b^2(1-(a/b)^{m+1})^2}{(b-a)^2} \to \frac{b^2}{(b-a)^2} \neq 0,$$

so the series diverges there, and it diverges for $|x| > 1/b^2$ too, where the terms grow without bound.

Checks

Take $a = 1$, $b = 2$. Then $c_m = 2^{m+1} - 1$, so $c_0, c_1, c_2 = 1, 3, 7$, and directly $\frac{1}{(1-x)(1-2x)} = (1 + x + x^2 + \dots)(1 + 2x + 4x^2 + \dots) = 1 + 3x + 7x^2 + \cdots$. The squares are $1, 9, 49$. On the other side, $\frac{1}{(1-x)(1-2x)(1-4x)} = 1 + 7x + 35x^2 + \cdots$, and multiplying by $1 + 2x$ gives $1 + 9x + 49x^2 + \cdots$, as it should. Numerically, with $a = 0.7$, $b = 1.3$ and $x = 0.3$, summing $c_m^2 x^m$ to many terms gives $4.163878\ldots$, which matches the closed form.

Common slips

- Giving the interval as $b|x| < 1$, copied from the first part. The new series has ratios a^2x, abx, b^2x, so the condition is $b^2|x| < 1$.
- Leaving c_m as a sum. The closed form $(b^{m+1} - a^{m+1})/(b - a)$ is what makes the rest of the question work.
- Sign errors in the coefficient of x in N. Expanding each product carefully, term by term, is quicker than trying to be clever.

Beyond STEP

The series $\sum c_m x^m$ is the *generating function* of the sequence c_m, and the question shows that if a sequence has a rational generating function, so does the sequence of its squares. This is a special case of the *Hadamard product* of two rational power series, which is always rational. The interval found here is an instance of the *radius of convergence*: a power series of a rational function converges up to the nearest pole, here $x = 1/b^2$. Generating functions are developed beautifully in Wilf's *generatingfunctionology*, and power series and their convergence are treated carefully in Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.